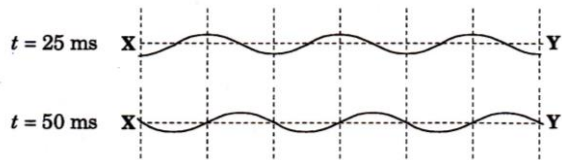
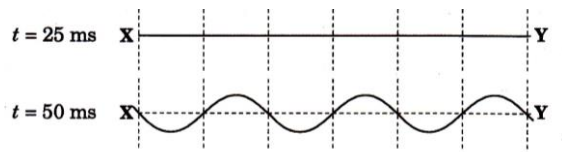


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	0.30 [m]	1			1		
		(ii)	$v = \frac{0.30 \text{ [m]}}{0.1 \text{ [s]}}$ ecf or $v = 10 \text{ [Hz]} \times 0.3 \text{ [m]}$ ecf or by implication (1) $v = 3.0 \text{ [m s}^{-1}\text{]} (1)$		2		2	2	
		(iii)	 1 mark for each time - graph drawn correctly		2		2	2	
	(b)	(i)	Wave reflected [at fixed point] (1) accept 2 waves travelling in opposite directions Stationary wave arises from superposition or interference (1)	1	1		2		
		(ii)	 1 mark for each time - graph drawn correctly	2			2	2	
			Question 4 total	4	5	0	9	6	0